Question 1 **[35 Marks]**

Note: *As for question (a) draw the diagram in the answer booklet provided. As for questions (b) and (c), modify the same program, program1.cpp.*

Given a C++ program named **program1.cpp** that contains four classes named `Course`, `Instructor`, `TextBook` and `Name`, respectively. Analyze the program and answer the following questions:

a. Draw the UML class diagram based on the implementation of all the classes in **program1.cpp**. (10 marks)

b. Define a C++ template for a class named `Array` to store an array of object items in the same program **program1.cpp**. The detail specifications of the class are as follows:
   i. Store the items in a `vector`.
   ii. Define a method named `add()` to add a new item (passed as a parameter) to the array.
   iii. Define a method named `getItem()` that returns an item based on the index passed as parameter to the method.
   iv. Define a method named `print()` to print all the items of the array.

(15 marks)

c. Test the template in the main function of **program1.cpp** by accomplishing the following tasks. See the expected output in Figure 1 (see next page).
   i. Create an array of instructors using the template and print all the items of the array.
   ii. Create an array of textbooks using the template and print all the items of the array.
   iii. Create a `Course` object with the following data and print the object.

      Title: **“Programming Technique II”**

      Instructor: the first item of the instructor array, created in c(i).

      Textbook: the second item of the textbook array, created in c(ii).

(10 marks)

```
LIST OF INSTRUCTORS:

First name: Ahmad
Last name: Kamal
Office number: N28-L305

First name: Mellisa
Last name: Wong
Office number: D06-105

First name: Ali
Last name: Hassan
Office number: N28A-L501

-------------------------------------------------------------
LIST OF TEXTBOOKS:

Title: Object-Oriented Programming Approach Using C++
Author: Gilberg
Publisher: Springer Publication

Title: Introduction to Functional Programming with TypeScript
Author: Samantha
Publisher: Oxford Press

-------------------------------------------------------------
COURSE INFORMATION:

Course name: Programming Technique II
Instructor Information:
First name: Mellisa
Last name: Wong
Office number: D06-105

Textbook Information:
Title: Object-Oriented Programming Approach Using C++
Author: Gilberg
Publisher: Springer Publication
```

Figure 1: Expected output of program1

Question 2 **[65 Marks]**

Note: *As for questions (a), (b) and (c), write your answer in the answer booklet provided. As for question (d) write the code in a C++ file (e.g., program2.cpp). Use only a single program for this question.*

Given a class diagram that models a student advisory system in Figure 2. Analyze the diagram and answer the following questions.

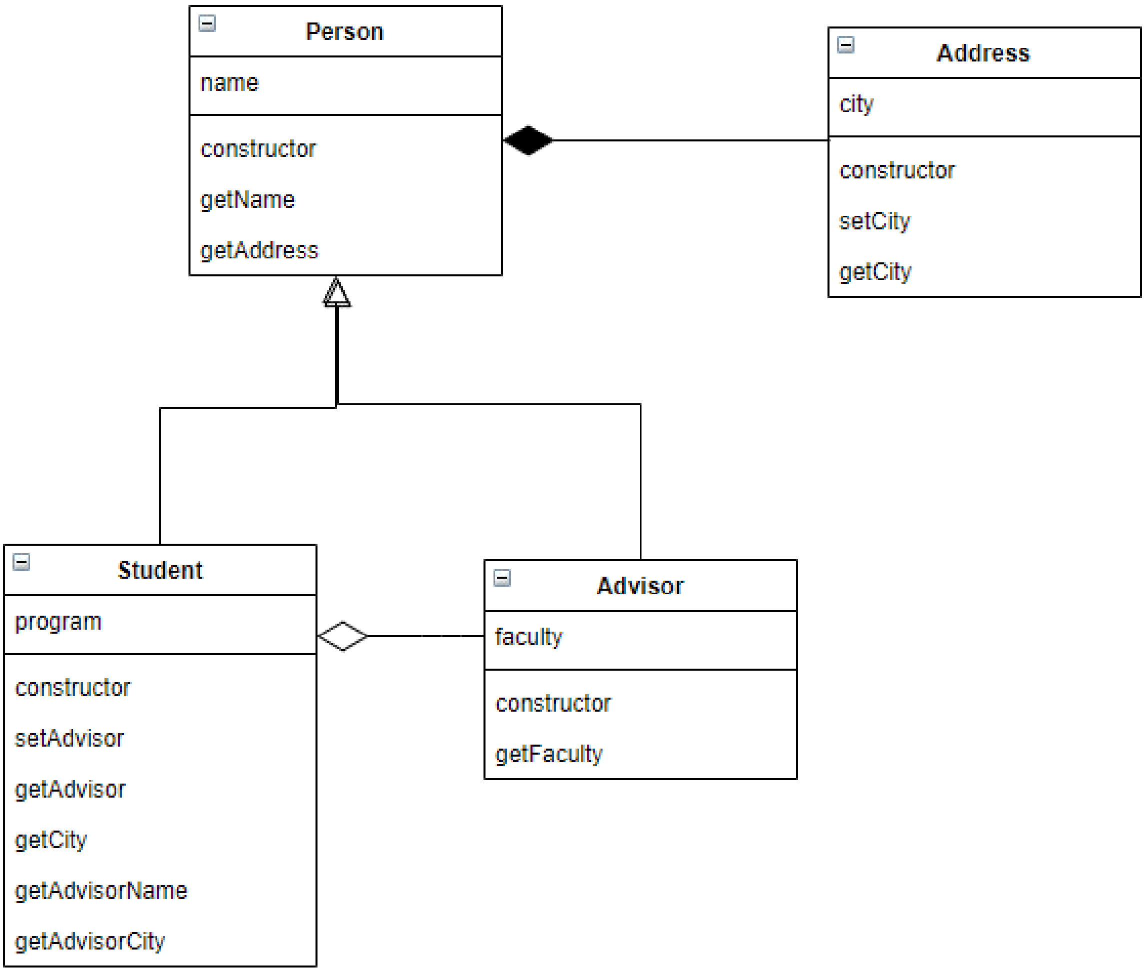

Figure 2: Class diagram for student advisory system

a. Determine the relationship between the following classes and justify your answers:

   i. `Student` and `Address`

   ii. `Student` and `Advisor`

   iii. `Advisor` and `Person`

(15 marks)

b. Can an advisor have more than one student under him or her? Justify your answer.

(5 marks)

c. Is it possible to define a method in the class `Advisor` to get the student under him or her? For example, `Advisor::getStudent()`. Justify your answer.

(5 marks)

d. Based on the class diagram, implement all classes in a C++ program. Define only the methods that are shown in the diagram (other methods are not required). The purpose of each method is as its name implies. Use inline style for all classes. Your program should have the followings:

i. Implementation of class `Address` (4 marks)

ii. Implementation of class `Person` (7 marks)

iii. Implementation of class `Advisor` (7 marks)

iv. Implementation of class `Student` (15 marks)

v. Test the classes in the main function by creating a student object, and assign it to an advisor. Finally print out the the information about the student including his/her advisor's information. See the expected output in Figure 3 (7 marks)

```
Student's name: Ali
Student's city: Skudai
Advisor's name: Dr. Abu
Advisor's city: Johor Bharu
Advisor's faculty: FC
```

Figure 3: Expected output of program2